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CAR OF ELEVATOR, AND ELEVATOR

Abstract

Provided are a car of an elevator, and an elevator that achieve weight reduction for a support frame. The support frame includes brackets at multiple locations on an upper edge, and supports a first pulley, a second pulley, and a door position detection part via the brackets. This eliminates the need to secure an area on a front face of the support frame for arranging the first pulley, the second pulley, and the door position detection part. Therefore, the height dimension of the support frame can be reduced to achieve weight reduction for the support frame.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefits of Japanese application no. 2024-036831, filed on Mar. 11, 2024, and Japanese application no. 2024-037821, filed on Mar. 12, 2024. The entirety of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a car of an elevator, and an elevator that include a car door device.

Description of Related Art

[0003] There is known a car of this type, which includes: (i) a door rail provided along an upper edge of an entrance of the car in the left-right direction, (ii) a car door supported to be slidable on the door rail, (iii) a first pulley and a second pulley provided to be spaced apart in the left-right direction above the door rail, (iv) a drive belt wound around the first pulley and the second pulley and operatively connected to the car door, and (v) a plate-shaped support frame supporting the door rail, the first pulley, and the second pulley on a front face (Japanese Patent Application Laid-Open No. 2011-514298).

[0004] However, due to the arrangement of the door rail, the first pulley (and a drive motor in the case where the drive motor is directly connected to the first pulley), and the second pulley on the front face of the support frame, the conventional car faces a problem that the height dimension of the support frame is large, which results in an increase in the weight of the support frame.

SUMMARY

[0005] In view of such circumstances, the disclosure provides a car of an elevator, and an elevator that achieve weight reduction for the support frame.

[0006] The following presents a simplified summary of the invention disclosed herein in order to provide a basic understanding of some aspects of the invention. This summary is not an extensive overview of the invention. It is intended to neither identify key or critical elements of the invention nor delineate the scope of the invention. Its sole purpose is to present some concepts of the invention in a simplified form as a prelude to the more detailed description that is presented later.

[0007] The car of the elevator, comprises: [0008] a door rail, supporting a car door to be slidable in a left-right direction; [0009] a first pulley and a second pulley provided to be spaced apart in the left-right direction above the door rail, and a drive belt operatively connected to the car door and being wound around the first pulley and the second pulley; [0010] a door position detection part, detecting that the car door is at a predetermined position; and [0011] a support frame, comprising the door rail on a front face and brackets at a plurality of locations on an upper edge, and the support frame supporting the first pulley, the second pulley, and the door position detection part via the brackets, [0012] wherein the car door comprises a roller that engages with an upper edge of the door rail from above and rotates, and [0013] the upper edge of the support frame is positioned lower than an uppermost portion of the roller.

[0014] The aforementioned car of the elevator can be configured such that further comprises: [0015] a drive motor, directly connected to the first pulley, [0016] wherein the drive motor is supported on the support frame via one of the brackets.

[0017] The aforementioned car of the elevator can be configured such that the brackets are attached to a rear face of a main body part of the support frame.

[0018] The aforementioned car of the elevator can be configured such that the support frame is a member that is long in the left-right direction and is arranged along a vertical plane, and [0019] the first pulley, the second pulley, and the door position detection part are arranged in an open space

between a front offset plane and a rear offset plane with reference to the vertical plane of the support frame, above the support frame.
[0020] The aforementioned car of the elevator can be configured such that further comprises:
[0021] a cover that covers a part or entirety of the open space from front, rear, and above.
[0022] The aforementioned car of the elevator can be configured such that the support frame comprises: [0023] a main body part; and [0024] a front protrusion part that protrudes forward from at least a part of an upper edge of the main body part, [0025] the bracket supporting the drive motor has an upper face in a part, and is attached to the support frame so that the upper face is at the same height position as an upper face of the front protrusion part, and [0026] the drive motor has a base face, and is attached to the bracket so that the base face straddles and contacts the upper face of the front protrusion part and the upper face of the bracket.
[0027] The aforementioned car of the elevator can be configured such that the drive motor is attached to the bracket so that a tip of the base face aligns with a tip of the upper face of the front protrusion part.
[0028] The aforementioned car of the elevator can be configured such that the support frame comprises: [0029] a main body part; and [0030] a front protrusion part that protrudes forward from at least a part of an upper edge of the main body part, and [0031] the brackets have an upper face in a part, and is attached to the support frame so that the upper face is at the same height position as an upper face of the front protrusion part.
[0032] The aforementioned car of the elevator can be configured such that the brackets are attached to the support frame using a fastener that is able to be fixed and released by a rotational operation from front of the support frame.
[0033] The aforementioned car of the elevator can be configured such that the car door comprises: [0034] a door panel; and [0035] a door hanger that is attached to an upper end portion of the door panel, [0036] the door hanger comprises an extension part where a portion of a main body part of the support frame protrudes and extends upward, and [0037] the car door is connected to the drive belt at the extension part.
[0038] The aforementioned car of the elevator can be configured such that the extension part comprises a detection piece for the door position detection part.
[0039] The aforementioned car of the elevator can be configured such that the support frame comprises a pair of brackets spaced apart in the left-right direction on a rear face, and [0040] each of the pair of brackets comprises: [0041] a main body part, attached to the support frame; and [0042] a rear protrusion part, protruding rearward from an upper portion of the main body part, [0043] wherein a lower end of the main body part and a tip of the rear protrusion part are grounded to function as a base of the support frame.
[0044] The aforementioned car of the elevator can be configured such that each of the pair of brackets has a length in an up-down direction that is longer than a length of the support frame in the up-down direction.
[0045] The aforementioned car of the elevator can be configured such that each of the pair of brackets is attached to the support frame so as to be changeable in position in the up-down direction.
[0046] The aforementioned car of the elevator can be configured such that each of the pair of brackets comprises an engaged part for a suspension tool at a location on the rear protrusion part that is behind a center of gravity of an integrated structure of the door rail, the drive motor, the first pulley, the second pulley, and the support frame.
[0047] The car of the elevator, comprises: [0048] a door rail, supporting a car door to be slidable in a left-right direction; [0049] a first pulley and a second pulley provided to be spaced apart in the left-right direction above the door rail, and a drive belt operatively connected to the car door and being wound around the first pulley and the second pulley; [0050] a door position detection part, detecting that the car door is at a predetermined position; and [0051] a support frame, comprising

the door rail on a front face and brackets at a plurality of locations on an upper edge, and the support frame supporting the first pulley, the second pulley, and the door position detection part via the brackets, wherein the brackets are attached to the support frame using a fastener that is able to be fixed and released by a rotational operation from front of the support frame.
[0052] The elevator comprises the car of the elevator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0053] The aforementioned features and the other features of the present invention will be clarified by the following description and figures illustrating the embodiments of the present invention.
[0054] FIG. **1** is a perspective view of the elevator.
[0055] FIG. **2**A is a front view of the car with the car door closed, as viewed from the landing side.
[0056] FIG. **2**B is a front view of the car with the car door opened.
[0057] FIG. **3** is a front view of the upper portion of the car with the car door closed.
[0058] FIG. **4** is a front view of the upper portion of the car with the car door opened.
[0059] FIG. **5**A is a cross-sectional view along line A in FIG. **3**.
[0060] FIG. **5**B is a cross-sectional view along line B in FIG. **3**.
[0061] FIG. **6**A is a front view of the upper portion of the car with the car door removed.
[0062] FIG. **6**B is a cross-sectional view along line C in FIG. **6**A.
[0063] FIG. **6**C is a cross-sectional view along line C in FIG. **6**A.
[0064] FIG. **6**D is an explanatory view related to another example of FIG. **6**C.
[0065] FIG. **7**A is a front view of portion D in FIG. **6**A.
[0066] FIG. **7**B is an exploded view of FIG. **7**A.
[0067] FIG. **8**A is a side view of FIG. **7**A.
[0068] FIG. **8**B is an exploded view of FIG. **8**A.
[0069] FIG. **9**A is a front view of portion E in FIG. **6**A.
[0070] FIG. **9**B is an exploded view of FIG. **9**A.
[0071] FIG. **10**A is a side view of FIG. **9**A.
[0072] FIG. **10**B is an exploded view of FIG. **10**A.
[0073] FIG. **11**A is a front view of portion F in FIG. **6**A.
[0074] FIG. **11**B is an exploded view of FIG. **11**A.
[0075] FIG. **12**A is a side view of a portion of FIG. **11**A.
[0076] FIG. **12**B is an exploded view of FIG. **12**A.
[0077] FIG. **13**A is a side view of another portion of FIG. **11**A.
[0078] FIG. **13**B is an exploded view of FIG. **13**A.
[0079] FIG. **14**A is a front view of portion G in FIG. **6**A.
[0080] FIG. **14**B is an exploded view of FIG. **14**A.
[0081] FIG. **15** is a front view of portion H in FIG. **6**A.
[0082] FIG. **16** is a side view of FIG. **15**.
[0083] FIG. **17** is a front view of the integrated structure with various components attached to the support frame.
[0084] FIG. **18**A is a front view of portion I in FIG. **17**.
[0085] FIG. **18**B is an exploded view of FIG. **18**A.
[0086] FIG. **19**A is a side view of FIG. **18**A.
[0087] FIG. **19**B is an exploded view of FIG. **19**A.
[0088] FIG. **20**A is a modification example of FIG. **6**A.
[0089] FIG. **20**B is another example of FIG. **20**A.
[0090] FIG. **21**A is a cross-sectional view along line I in FIG. **20**A.

[0091] FIG. **21**B is a cross-sectional view along line J in FIG. **20**B.
[0092] FIG. **22**A is an explanatory view when the bracket is used as a base.
[0093] FIG. **22**B is an explanatory view when the bracket is used as a base.
[0094] FIG. **22**C is an explanatory view when the bracket is used as a base.
[0095] FIG. **22**D is an explanatory view when the bracket is used as a base.
[0096] FIG. **23** is an explanatory view of the installation work.

DESCRIPTION OF THE EMBODIMENTS

[0097] The following describes an embodiment of the elevator according to the disclosure. In the following description, when a person standing on the floor looks at the car (the state of FIG. **2**A and FIG. **2**B), the left side is referred to as "left," the right side is referred to as "right," the horizontal direction from left to right or from right to left is referred to as "left-right direction," and the left-right direction is denoted as X in appropriate figures. Additionally, when a person standing on the floor looks at the car (the state of FIG. **2**A and FIG. **2**B), the side close to the person is referred to as "front," the side far from the person is referred to as "back" or "rear," the horizontal direction from front to rear or from rear to front is referred to as "front-rear direction," and the front-rear direction is denoted as Y in appropriate figures. The left-right direction and the front-rear direction are two perpendicular directions in the horizontal plane. Furthermore, the up-down direction is synonymous with the vertical direction, and is perpendicular to both the left-right direction and the front-rear direction.
[0098] As shown in FIG. **1**, an elevator **1** includes a hoistway **2** and a car **3**. The hoistway **2** extends in the up-down direction within a building that has multiple floors. The car **3** ascends and descends within the hoistway **2** by the operation of a drive mechanism, and stops at a designated floor in response to the drive mechanism stopping the operation.
[0099] The car **3** includes a car door device **30**. The car door device **30** includes car doors **31**, a door support structure **32**, and a door opening/closing device **35**. The car doors **31** move in the left-right direction to open and close an opening (car entrance) on the front face of the car **3**, allowing a passenger to enter and exit. The door support structure **32** is a structure that supports the car doors **31** while allowing the car doors **31** to open and close. The door opening/closing device **35** is a device for opening and closing the car doors **31**.
[0100] On the other hand, a landing **20** on each floor includes a three-sided frame **21** and a landing door device **22**. The three-sided frame **21** is formed in a wall surface, with the interior of the frame forming an opening corresponding in size to the opening of the car **3**. The landing door device **22** includes landing doors **23** and a door support structure **24**. The landing doors **23** move in the left-right direction to open and close the opening (landing entrance) within the three-sided frame **21**, allowing a passenger to enter and exit. The landing doors **23** are normally closed by an elastic force, and are opened following the opening of the car doors **31** of the car **3** stopped at the floor and are closed along with the closing of the car doors **31**. The door support structure **24** is a structure that supports the landing doors **23** while allowing the landing doors **23** to open and close.
[0101] As shown in FIG. **2**A and FIG. **2**B, the car door **31** includes a door panel **310** and a door hanger **311**. The door panel **310** is arranged along the vertical plane, and is a main body serving as the door of the car door **31** for opening and closing the opening **3***a* of the car **3**. The door hanger **311** is integrally attached to the upper end portion of the door panel **310**. The door hanger **311** includes a pair of rollers **312**. The pair of rollers **312** are provided at the upper end portion of the door hanger **311**, and spaced apart in the left-right direction.
[0102] The door support structure **32** includes a door rail **33** and a sill **34**. The door rail **33** is a plate-shaped member that is straight, long in the left-right direction, flat, and has a rectangular shape. The door rail **33** is provided in the left-right direction on the upper portion of the front face of the car **3**, along the upper edge of the opening **3***a* of the car **3**, with both ends extending to the left and right beyond the upper edge of the opening **3***a*. The car doors **31** are suspended from the door rail **33** in a state where the car doors **31** are slidable in the left-right direction along the door

rail **33**, as the rollers **312** of the door hanger **311** engage with the upper edge of the door rail **33** from above and rotate. The door rail **33** is a structure that supports the upper portions of the car doors **31** while allowing the car doors **31** to open and close. On the other hand, the sill **34** is provided in the left-right direction on the lower portion of the front face of the car **3**, along the lower edge of the opening **3***a* of the car **3**, with both ends extending to the left and right beyond the lower edge of the opening **3***a*. The sill **34** is a structure that supports the lower portions of the car doors **31** while allowing the car doors **31** to open and close.

[0103] As shown in FIG. **3** and FIG. **4**, the door opening/closing device **35** includes a drive motor **350**, a drive pulley **351**, an idler pulley **352**, and a drive belt **353**. The drive motor **350** is provided on either the left or right end side of the upper portion of the front face of the car **3**. The drive pulley **351** is provided coaxially and integrally with the drive shaft of the drive motor **350** and is directly connected to the drive motor **350**. The idler pulley **352** is provided on the other end side of the upper portion of the front face of the car **3**. The drive belt **353** is an endless belt wound around the drive pulley **351** and the idler pulley **352**. The drive belt **353** is provided in the left-right direction in parallel to the door rail **33**.

[0104] In response to the drive motor **350** operating, the drive pulley **351** rotates, and accordingly, the drive belt **353** circulates. As a result, the two car doors **31** (hatched portions) on the left and right reciprocate along the door rail **33** (hatched portion) to open and close. One car door **31** is connected to one side of the drive belt **353**. The other car door **31** is connected to the other side of the drive belt **353**. Thus, the pair of car doors **31** move in opposite directions to each other. In other words, the car door device **30** and the landing door device **22** are of the center-opening type that the opening **3***a* of the car **3** opens from the center.

[0105] Here, the connecting structure between the car door **31** and the drive belt **353** will be described in detail. The door hanger **311** of the car door **31** includes a pair of first members and a second member. The pair of first members and the second member are plate-shaped members. The pair of first members are spaced apart in the left-right direction, and each is integrally attached to the door panel **310** at the upper end portion of the door panel **310**. The second member connects the pair of first members. In one car door **31**, the door hanger **311** includes an extension part **311***a* extending upward. In the other car door **31**, the door hanger **311** includes an extension part **311***b* extending upward. The extension part **311***a* and the extension part **311***b* are formed on the left half portion or the right half portion divided by the center line in the left-right direction of the first member. The first member is manufactured by cutting out the entire shape from a single plate material. Therefore, the extension part **311***a* and the extension part **311***b* are each a portion of the main body part of the first member which protrudes upward, and are integrally provided with the main body part.

[0106] As shown in FIG. **5**A and FIG. **5**B, the upper end portions of the extension part **311***a* and the extension part **311***b* are bent. The bent portion is a portion that protrudes rearward (or forward), and is a portion arranged along the horizontal plane. One side of the drive belt **353** is fixed using a fastener in a state of being in contact with the lower face (or upper face) of the bent portion of the extension part **311***a*. The other side of the drive belt **353** is fixed using a fastener in a state of being in contact with the upper face (or lower face) of the bent portion of the extension part **311***b*.

[0107] The roller **312** is rotatably attached to the inner face (of the first member) of the door hanger **311**. The roller **313** is rotatably attached to the inner face (of the second member) of the door hanger **311**. The roller **312** is in contact with the upper edge of the door rail **33**. The roller **313** is in contact with the lower edge of the door rail **33**. The door rail **33** is attached to the front face of a support frame **4** via legs **33***a*. The door rail **33** is provided in parallel to the support frame **4** on the front face of the support frame **4**. The leg **33***a* is a columnar member, and multiple legs **33***a* are provided to be spaced apart appropriately in the left-right direction and attached to the front face of the support frame **4**.

[0108] In one car door **31**, a switch operating member **314** is attached to the inner face (of the first member) of the door hanger **311**. The switch operating member **314** includes a roller and turns on a switch **354** when the car door **31** is in a closed state. In the other car door **31**, (the bent portion of) the extension part **311***b* of the door hanger **311** includes a detection piece **311***c*. The detection piece **311***c* is a detection piece for a limit switch **355** and a limit switch **356**.

[0109] As shown in FIG. **6**A to FIG. **6**D, the support frame **4** is a plate-shaped member that is straight, long in the left-right direction, flat, and has a rectangular shape. The support frame **4** is provided in the left-right direction on the upper portion of the front face of the car **3**, along the upper edge of the opening **3***a* of the car **3**, with both ends extending to the left and right beyond the upper edge of the opening **3***a*. The support frame **4** is a member for attaching and supporting various components which will be described later. The support frame **4** has rigidity that does not easily deform and is, for example, a metal member including iron, steel, stainless steel, or aluminum as the material.

[0110] The support frame **4** includes a main body part **40** and a front protrusion part **41**. The main body part **40** is a portion that occupies the majority of the entire support frame **4**, and is a plate-shaped portion that is straight, long in the left-right direction, flat, rectangular, and arranged along the vertical plane. The front protrusion part **41** is a portion that protrudes forward from the upper edge of the main body part **40**, and is a plate-shaped portion that is straight, long in the left-right direction, flat, rectangular, and arranged along the horizontal plane. The front protrusion part **41** has an upper face **41***a*. The upper face **41***a* is a horizontal face. The main body part **40** and the front protrusion part **41** are integrally provided. As an example, the main body part **40** and the front protrusion part **41** are formed by bending the upper end portion of a single plate material forward.

[0111] The support frame **4** has a smaller height dimension compared to the support frames in related art. Thus, the upper edge of the support frame **4** (the upper edge of the main body part **40** and the edge related to the upper face **41***a* of the front protrusion part **41**; hereinafter, the same when referred to as “the upper edge of the support frame”) is positioned lower than the uppermost portion of the roller **312** with a height difference H.

[0112] The support frame **4** is attached to the upper portion of the front face of the car **3**. As an example, the support frame **4** is attached to the upper portion of the front face of the car **3** via brackets **50** and appropriate brackets **50**A (FIG. **6**C). The upper portion of the front face of the car **3** includes a recessed part **3***b* that is recessed in the depth direction, and the support frame **4** is arranged in front of the recessed part **3***b*. Thus, the support frame **4** faces the back face of the recessed part **3***b* with a gap, and an open space S is formed above the support frame **4**. The open space S is a space between the front offset plane and the rear offset plane with reference to the vertical plane (of the main body part **40**) of the support frame **4**. The open space S becomes the arrangement space for various components which will be described later. The brackets **50** are attached to both end portions of the support frame **4**. The brackets **50**A are combined with the brackets **50** and attached to the outer face of the car **3**. As a result, the support frame **4** is fixed to the upper portion of the front face of the car **3**.

[0113] As another example, the support frame **4** is attached to the upper portion of the front face of the car **3** via front frames **3**A of the car **3** (FIG. **6**D). The front frames **3**A are provided as a pair on both sides of the car **3**, and the support frame **4** is arranged straddling the pair of front frames **3**A. Thus, the support frame **4** faces the front face of the car **3** with a gap, and a similar open space S is formed above the support frame **4**. The support frame **4** is attached to the pair of front frames **3**A at both ends. As a result, the support frame **4** is fixed to the upper portion of the front face of the car **3**.

[0114] Various components such as the drive motor **350** (and the drive pulley **351**), the idler pulley **352**, the switch **354**, the limit switch **355**, and the limit switch **356** of the door opening/closing device **35** are attached to appropriate locations of the support frame **4**. In addition, a control box **37** is attached to an appropriate location of the support frame **4** as another component. These various components are attached to the support frame **4** via brackets. The brackets are provided at multiple locations on the upper edge of the support frame **4**, and extend and protrude upward, rearward, or forward from the upper edge of the support frame **4**. Thus, the various components are arranged in the open space S. The brackets supporting the switch **354**, the limit switch **355**, and the limit switch **356** are arranged between the bracket supporting the drive motor **350** (and the drive pulley **351**) and the bracket supporting the idler pulley **352**, but of course, these brackets are provided so as not to interfere with the drive belt **353**. The brackets have rigidity that does not easily deform and are, for example, metal members including iron, steel, stainless steel, or aluminum as the material.

[0115] As shown in FIG. **7**A, FIG. **7**B, FIG. **8**A, and FIG. **8**B, a bracket **51** supporting the drive motor **350** and the drive pulley **351** is attached to the rear face (back face) (of the main body part **40**) of the support frame **4**. The bracket **51** is detachably attached to (the main body part **40** of) the support frame **4** using a fastener **61**. The fastener **61** is a bolt and nut fastener. The fastener **61** is a combination of two bolts **610** and one plate nut (a plate with two nuts welded thereto) **611** (including appropriate washers, spring washers, etc. as needed). (The upper end portion of) (the main body part **40** of) the support frame **4** has a round hole as the insertion hole for the shaft portion of the bolt **610**. (The lower end portion of) the bracket **51** has a notched hole **51***a* that opens at the lower edge as the insertion hole for the shaft portion of the bolt **610**. The bolt **610** is inserted into the insertion holes of the support frame **4** and the bracket **51** from the front of the support frame **4** and is screwed into the plate nut **611**. Thus, the fastener **61** can be fixed and released by a rotational operation from the front of the support frame **4**. Then, the insertion hole of the bracket **51** is the notched hole **51***a*. Thus, the bracket **51** is lifted upward and removed in a state where the fastener **61** is loosened.

[0116] The bracket **51** is formed by bending a plate material. The bent portion is a portion that protrudes rearward, and is a portion arranged along the horizontal plane. The bent portion has an upper face **51***b*. The upper face **51***b* is a face visible from the front of the support frame **4** and is a horizontal face. The bracket **51** is attached to the support frame **4** so that the upper face **51***b* is close to the upper face **41***a* (of the front protrusion part **41**) of the support frame **4** and is at the same height position as (flush with) the upper face **41***a*.

[0117] The drive motor **350** has a base face (lower face) **350***a* and is attached to the bracket **51** so that the base face **350***a* is in contact with the upper edge of the support frame **4**. Then, the drive motor **350** is attached to the bracket **51** so that the base face **350***a* straddles and contacts the upper face **41***a* of the support frame **4** and the upper face **51***b* of the bracket **51**.

[0118] The drive motor **350** is attached to the bracket **51** in a manner that allows position adjustment in the front-rear direction (to the extent of the gap between the shaft portion of the bolt and the insertion hole). Then, the drive motor **350** is attached to the bracket **51** so that the tip of the base face **350***a* aligns with the tip of the upper face **41***a* of the support frame **4**.

[0119] As shown in FIG. **9**A, FIG. **9**B, FIG. **10**A, and FIG. **10**B, a bracket **52** supporting the idler pulley **352** is attached to the rear face (of the main body part **40**) of the support frame **4**. The bracket **52** is detachably attached to (the main body part **40** of) the support frame **4** using a fastener **62**. The fastener **62** is a bolt and nut fastener. The fastener **62** is a combination of two square root bolts **620** and two nuts **621** (including appropriate washers, spring washers, etc. as needed). (The upper end portion of) (the main body part **40** of) the support frame **4** has a square hole **42***a* as the insertion hole for the shaft portion of the square root bolt **620**. (The lower end portion of) the bracket **52** has a square hole **52***a* as the insertion hole for the shaft portion of the square root bolt **620**. The square root bolt **620** is inserted into the insertion holes of the bracket **52** and the support frame **4** from the rear of the support frame **4**, and the nut **621** is screwed on from the front of the support frame **4**. Thus, the fastener **62** can be fixed and released by a rotational operation from the front of the support frame **4**. Then, the square root portion with a square cross-section of the square root bolt **620** fits into the square hole **42***a* and the square hole **52***a*. Thus, the movement and rotation of the bracket **52** relative to the support frame **4** in the vertical plane are restricted.

[0120] The bracket **52** is formed using a plate material. The idler pulley **352** is rotatably attached to the bracket **52**. Although not shown, the idler pulley **352** is attached to the bracket **52** in a manner that allows position adjustment in the left-right direction. By adjusting the position of the idler pulley **352** in the left-right direction, the tension of the drive belt **353** is adjusted.

[0121] As shown in FIG. **11**A, FIG. **11**B, FIG. **12**A, and FIG. **12**B, a bracket **54** supporting the switch **354** is attached to the rear face (of the main body part **40**) of the support frame **4**. The bracket **54** is detachably attached to (the main body part **40** of) the support frame **4** using a fastener **64**. The fastener **64** is a bolt and nut fastener. The fastener **64** is a combination of two bolts **640** and one plate nut **641** (including appropriate washers, spring washers, etc. as needed). (The upper end portion of) (the main body part **40** of) the support frame **4** has an elongated hole **44***a* as the insertion hole for the shaft portion of the bolt **640**. (The lower end portion of) the bracket **54** has an elongated hole **54***a* as the insertion hole for the shaft portion of the bolt **640**. The elongated hole **44***a* is an elongated hole that is long in either the left-right direction or the up-down direction (in the example shown, the up-down direction). The elongated hole **54***a* is an elongated hole that is long in the other direction (in the example shown, the left-right direction). The bolt **640** is inserted into the insertion holes of the support frame **4** and the bracket **54** from the front of the support frame **4** and is screwed into the plate nut **641**. Thus, the fastener **64** can be fixed and released by a rotational operation from the front of the support frame **4**. Then, the insertion holes of the support frame **4** and the bracket **54** are the elongated hole **44***a* and the elongated hole **54***a* that are long in directions perpendicular to each other (the left-right direction and the up-down direction). This allows the position of the bracket **54** to be adjusted in the left-right direction and the up-down direction relative to the support frame **4**.

[0122] The bracket **54** is formed by bending a plate material at multiple locations. The bracket **54** is formed so that the switch **354** is arranged behind the drive belt **353**. The bent portion formed in the middle section of the bracket **54** is a portion that protrudes rearward and is a portion arranged along the horizontal plane. The bent portion has an upper face **54***b*. The upper face **54***b* is a face visible from the front of the support frame **4** and is a horizontal face. The bracket **54** is attached to the support frame **4** so that the upper face **54***b* is close to the upper face **41***a* (of the front protrusion part **41**) of the support frame **4** and is at the same height position as (flush with) the upper face **41***a*.

[0123] The switch **354** is attached to the bracket **54** in a manner that allows position adjustment in the front-rear direction and the left-right direction (to the extent of the gap between the shaft portion of the bolt and the insertion hole). The position adjustment of the switch **354** in the up-down direction is performed by adjusting the position of the bracket **54** in the up-down direction so that the upper face **54***b* of the bracket **54** is at the same height position as the upper face **41***a* of the support frame **4**.

[0124] The switch **354** is a switch detecting that the car door **31** is at the closed position. The switch **354** includes a switch piece **354***a*. The switch piece **354***a* is pivotally provided on the lower face of the main body part of the switch **354**, and when the car door **31** is in the closed state, the switch operating member **314** of the car door **31** comes into contact with the switch piece **354***a*, thereby pushing the switch piece **354***a* upward and turning the switch **354** ON. The switch **354** is in the form of an interlock switch that constitutes a part of the safety circuit of the drive device of the elevator **1**.

[0125] As shown in FIG. **11**A, FIG. **11**B, FIG. **13**A, and FIG. **13**B, a bracket **55** supporting the limit switch **355** is attached to the rear face (of the main body part **40**) of the support frame **4**. The bracket **55** is detachably attached to (the main body part **40** of) the support frame **4** using a fastener **65**. The fastener **65** is a bolt and nut fastener. The fastener **65** is a combination of two bolts **650** and one plate nut **651** (including appropriate washers, spring washers, etc. as needed). (The upper end portion of) (the main body part **40** of) the support frame **4** has an elongated hole **45***a* as the insertion hole for the shaft portion of the bolt **650**. (The lower end portion of) the bracket **55** has an elongated hole **55***a* as the insertion hole for the shaft portion of the bolt **650**. The elongated hole **45***a* is an elongated hole that is long in either the left-right direction or the up-down direction (in

the example shown, the up-down direction). The elongated hole **55***a* is an elongated hole that is long in the other direction (in the example shown, the left-right direction). The bolt **650** is inserted into the insertion holes of the support frame **4** and the bracket **55** from the front of the support frame **4** and is screwed into the plate nut **651**. Thus, the fastener **65** can be fixed and released by a rotational operation from the front of the support frame **4**. Then, the insertion holes of the support frame **4** and the bracket **55** are the elongated hole **45***a* and the elongated hole **55***a* that are long in directions perpendicular to each other (the left-right direction and the up-down direction). This allows the position of the bracket **55** to be adjusted in the left-right direction and the up-down direction relative to the support frame **4**.
[0126] The bracket **55** is formed by bending a plate material at multiple locations. The bent portion formed in the middle section of the bracket **55** is a portion that protrudes rearward and is a portion arranged along the horizontal plane. The bent portion has an upper face **55***b*. The upper face **55***b* is a face visible from the front of the support frame **4** and is a horizontal face. The bracket **55** is attached to the support frame **4** so that the upper face **55***b* is close to the upper face **41***a* (of the front protrusion part **41**) of the support frame **4** and is at the same height position as (flush with) the upper face **41***a*.
[0127] The limit switch **355** is attached to the bracket **55** in a manner that allows position adjustment in the front-rear direction and the left-right direction (to the extent of the gap between the shaft portion of an unillustrated bolt and the insertion hole). The position adjustment of the limit switch **355** in the up-down direction is performed by adjusting the position of the bracket **55** in the up-down direction so that the upper face **55***b* of the bracket **55** is at the same height position as the upper face **41***a* of the support frame **4**.
[0128] The limit switch **355** is a switch detecting that the car door **31** is at the closed position. As an example, the limit switch **355** is a proximity sensor. Since the role of the limit switch **355** may also be fulfilled by the switch **354**, it is possible to omit the limit switch **355**.
[0129] As shown in FIG. **14**A and FIG. **14**B, the limit switch **356** is a switch detecting that the car door **31** is at the opened position. The limit switch **356**, the bracket **56** supporting the limit switch **356**, the fastener **66**, and the fixing structure are substantially the same as the limit switch **355**, the bracket **55**, the fastener **65**, and the fixing structure shown in FIG. **11**A, FIG. **11**B, FIG. **13**A, and FIG. **13**B, although they are in a line-symmetrical arrangement relationship. Therefore, the description of the configuration is omitted as the description is the same.
[0130] As shown in FIG. **15** and FIG. **16**, a bracket **57** supporting the control box **37** is attached to the rear face (of the main body part **40**) of the support frame **4**. The bracket **57** is detachably attached to (the main body part **40** of) the support frame **4** using a fastener **67**. The fastener **67** includes two bolts **670** (including appropriate washers, spring washers, etc., as needed). (The upper end portion of) (the main body part **40** of) the support frame **4** has a round hole as the insertion hole for the shaft portion of the bolt **670**. (The lower end portion of) the bracket **57** has a threaded hole **57***a* for the bolt **670**. The bolt **670** is inserted into the insertion hole of the support frame **4** from the front of the support frame **4** and is screwed into the threaded hole **57***a*. Thus, the fastener **67** can be fixed and released by a rotational operation from the front of the support frame **4**.
[0131] The control box **37** houses a sub-control part for the control part (main control part) of the entire elevator **1**. Alternatively, the control box **37** houses a relay part for the control part of the entire elevator **1**. In some cases, the control box **37** may not be provided on the support frame **4**.
[0132] FIG. **17** shows an integrated structure (hereinafter simply referred to as "integrated structure") of the door rail **33**, the drive motor **350**, the drive pulley **351**, the idler pulley **352**, the switches **354** to **356**, and the support frame **4**, with everything attached except for the control box **37**.
[0133] (The support frame **4** of) the integrated structure includes a pair of brackets **50** spaced apart in the left-right direction on the rear face. As an example, the bracket **50** is arranged at each end portion of the support frame **4**. The brackets **50** have been described with reference to FIG. **6**A to

FIG. **6**D.

[0134] As shown in FIG. **18**A, FIG. **18**B, FIG. **19**A, and FIG. **19**B, the bracket **50** includes a main body part **500** and a rear protrusion part **501**. The main body part **500** is a plate-shaped portion that is straight, flat, rectangular, and arranged along the vertical plane. The rear protrusion part **501** is a portion that protrudes rearward from the upper edge of the main body part **500**, and is a plate-shaped portion that is straight, flat, rectangular, and arranged along the horizontal plane. The main body part **500** and the rear protrusion part **501** are integrally formed. As an example, the main body part **500** and the rear protrusion part **501** are formed in an inverted L-shape by bending the upper portion of a single plate material rearward.

[0135] (The main body part **500** of) the bracket **50** has a length H**50** in the up-down direction that is longer than a length H**4** of (the main body part **40** of) the support frame **4** in the up-down direction. Thus, the bracket **50** protrudes and extends downward from the lower edge of the support frame **4**, and a lower end **50***a* (of the main body part **500**) of the bracket **50** is positioned at a height lower than the lower edge of the support frame **4**. Additionally, the bracket **50** protrudes and extends upward from the upper edge of the support frame **4**, and a tip **50***b* (of the rear protrusion part **501**) of the bracket **50** is positioned at a height higher than the upper edge of the support frame **4**.

[0136] The bracket **50** is attached to the rear face (of the main body part **40**) of the support frame **4**. The bracket **50** is detachably attached to (the main body part **40** of) the support frame **4** using a fastener **60**. The fastener **60** is a bolt and nut fastener. The fastener **60** is a combination of two bolts **600** and two nuts **601** (including appropriate washers, spring washers, etc., as needed). (The upper end portion of) (the main body part **40** of) the support frame **4** has through holes **40***a* as the insertion holes for the shaft portions of the bolts **600**. Multiple through holes **40***a* are provided in a row in the left-right direction, and two appropriate through holes **40***a* are selected. (The main body part **500** of) the bracket **50** has slide elongated holes **500***a* that are long in the up-down direction as the insertion holes for the shaft portions of the bolts **600**. Two slide elongated holes **500***a* are provided in a row in the left-right direction. The two slide elongated holes **500***a* are in a left-right symmetrical arrangement relationship centered on the center line of the width of (the main body part **500** of) the bracket **50** in the left-right direction. Each of the two slide elongated holes **500***a* is formed to be longer than the length H**4** of the support frame **4** in the up-down direction. The bolt **600** is inserted into the insertion holes of the support frame **4** and the bracket **50** from the front of the support frame **4** and is screwed into the nut **601**. Thus, the fastener **60** can be fixed and released by a rotational operation from the front of the support frame **4**. Then, the insertion hole of the bracket **50** is the slide elongated hole **500***a*. This allows the bracket **50** to be repositioned and slidable in the up-down direction in a state where the fastener **60** is loosened.

[0137] The bracket **50** includes an eye bolt **502** as an engaged part for a suspension tool. The eye bolt **502** is provided at a location on the rear protrusion part **501** that is behind the center of gravity of the integrated structure. The eye bolt **502** is provided to protrude upward from the upper face of the rear protrusion part **501**.

[0138] The bracket **50** includes a fastener **60**A at the end portion of the rear protrusion part **501**. The fastener **60**A is a bolt and nut fastener, and is a combination of one bolt **600**A and one nut **601**A. The fastener **60**A is for fixing the bracket **50** and the bracket **50**A (FIG. **6**C).

[0139] (The lower end portion of) (the main body part **40** of) the support frame **4** has notched holes **40***b* that open at the lower edge. Multiple notched holes **40***b* are appropriately spaced apart in the left-right direction along the entire length of the support frame **4**. The notched hole **40***b* is used as the insertion hole for the shaft portion of the bolt **600**A of the fastener **60**A that is for fixing the support frame **4** and the separate bracket **50**A (FIG. **6**C).

[0140] As shown in FIG. **20**A, FIG. **20**B, FIG. **21**A, and FIG. **21**B, the support frame **4** includes a cover **38**. The cover **38** covers a part or entirety of the open space S in the left-right direction from the front, rear, and above. Thus, the switch **354**, the limit switch **355**, the limit switch **356**, and the idler pulley **352** are covered by the cover **38** (FIG. **20**A). Alternatively, the drive motor **350**, the drive pulley **351**, the switch **354**, the limit switch **355**, the limit switch **356**, and the idler pulley **352** are covered by the cover **38** (FIG. **20**B). In both cases, the upper portion of the car door **31** (the upper portion of the door hanger **311**) and the drive belt **353** are also covered by the cover **38**.

[0141] The bracket **58** supporting the cover **38**, the fastener, and the fixing structure are the same as the bracket **57**, the fastener **67**, and the fixing structure shown in FIG. **15** and FIG. **16**. Therefore, the description of the configuration is omitted as the description is the same.

[0142] As shown in FIG. **3** and FIG. **4**, the car door device **30** includes a landing door engagement device **39**. The landing door engagement device **39** is a device that, when the car doors **31** open (and close), engages with an engaged body (for example, roller) **230** of the landing door **23**, thereby causing the landing doors **23** to open (and close) in accordance with the opening (and closing) of the car doors **31**. The landing door engagement device **39** includes, as components on the side of the car door **31**, a first engagement body **390**, a second engagement body **391**, and a link **392**, and as a component on the entrance side of the car **3**, a cam **393**.

[0143] The first engagement body **390** is fixed. The second engagement body **391** is provided to be movable between a first state (FIG. **3**) and a second state (FIG. **4**) where the second engagement body **391** is closer to the first engagement body **390** and the gap (width) between the second engagement body **391** and the first engagement body **390** is narrower than in the first state. The link **392** is operatively connected to the second engagement body **391** at one end portion. As shown in FIG. **3**, when the car doors **31** are in a closed state, (the roller at) the other end portion of the link **392** comes into contact with the cam **393**, thereby displacing the second engagement body **391** from the second state to the first state. Then, as shown in FIG. **4**, when the car doors **31** open and the other end portion of the link **392** moves away from the cam **393**, the second engagement body **391** is displaced from the first state to the second state due to its own weight.

[0144] As shown in FIG. **11**A and FIG. **11**B, the cam **393** is attached to the front face of the support frame **4** via a bracket **394** at an intermediate position in the left-right direction of the support frame **4**.

[0145] The configuration of the elevator **1** according to this embodiment is as described above. According to the elevator **1** of this embodiment, the support frame **4** includes the brackets **51** to **57** at multiple locations along the upper edge, and supports various components such as the drive motor **350** (and the drive pulley **351**), the idler pulley **352**, the switch **354**, the limit switch **355**, the limit switch **356**, and the control box **37** via the brackets **51** to **57**. Thus, there is no need to secure an area on the front face of the support frame **4** to arrange these various components. Therefore, it is possible to reduce the height dimension of the support frame **4**, thereby achieving weight reduction for the support frame **4**, and consequently, reducing the manufacturing cost of the car **3**, improving the work efficiency in installation work, and reducing the operating cost.

[0146] Furthermore, according to the elevator **1** of this embodiment, the drive pulley **351** is directly connected to the drive motor **350**, and the drive motor **350** is supported on the support frame **4** via the bracket **51**. Additionally, the drive motor **350** has the base face **350***a* and is attached to the bracket **51** so that the base face **350***a* is in contact with the upper edge of the support frame **4**. As a result, in the system (direct drive system) where the drive belt **353** is directly driven by the drive motor **350**, it is possible to bring the height position of the drive belt **353** closer to the roller **312** of the car door **31**, and it is possible to suppress the eccentric load (eccentric torque) associated with driving from acting on the car doors **31**. Therefore, it is possible to improve the movement stability of the car doors **31** in the left-right direction, and to open and close the car doors **31** smoothly and quietly.

[0147] Moreover, according to the elevator **1** of this embodiment, the upper edge of the support frame **4** is positioned lower than the uppermost portion of the roller **312**. Thus, it is possible to bring the height position of the drive belt **353** even closer to the roller **312**, and it is possible to further suppress the eccentric load (eccentric torque) associated with driving from acting on the car doors **31**. Therefore, it is possible to further improve the movement stability of the car doors **31** in the left-right direction, and to open and close the car doors **31** more smoothly and more quietly. Furthermore, it is possible to further reduce the height dimension of the support frame **4**, thereby achieving further weight reduction for the support frame **4**.

[0148] Further, according to the elevator **1** of this embodiment, the support frame **4** is a member that is long in the left-right direction and arranged along the vertical plane, and various components are arranged in the open space S between the front offset plane and the rear offset plane with reference to the vertical plane of the support frame **4**, above the support frame **4**. Therefore, it is possible to neatly configure the area above the support frame **4**.

[0149] Additionally, according to the elevator **1** of this embodiment, the cover **38** is provided to cover a part or entirety of the open space S from the front, rear, and above. Therefore, it is possible to prevent dust or the like from accumulating on various components. Moreover, as the area above the support frame **4** is neatly configured, it is possible to make the cover **38** compact.

[0150] Furthermore, according to the elevator **1** of this embodiment, the support frame **4** is a member that is long in the left-right direction, and includes the main body part **40** and the front protrusion part **41** that protrudes forward from the upper edge of the main body part **40**. Therefore, it is possible to obtain high rigidity even with the downsized support frame **4**. Additionally, there is no protrusion part toward the rear, and the back face of the support frame **4** is a flat face. Therefore, the support frame **4** can be easily attached to the structure of the car **3**.

[0151] Moreover, according to the elevator **1** of this embodiment, the brackets **51** to **58** are attached to the rear face of the main body part **40** of the support frame **4**. Therefore, it is possible to neatly configure the front face of the support frame **4**.

[0152] Further, according to the elevator **1** of this embodiment, the bracket **51** supporting the drive motor **350** has the upper face **51***b* in a part, and is attached to the support frame **4** so that the upper face **51***b* is at the same height position as the upper face **41***a* of the front protrusion part **41** of the support frame **4**. The drive motor **350** has the base face **350***a* and is attached to the bracket **51** so that the base face **350***a* straddles and contacts the upper face **41***a* of the front protrusion part **41** and the upper face **51***b* of the bracket **51**. Thus, the drive motor **350** is constrained by the two upper faces **41***a* and **51***b* in the rotational direction around the drive shaft. Therefore, even if rotational torque is generated in the drive motor **350** along with the driving of the drive motor **350**, it is possible to prevent the drive motor **350** from causing misalignment in the rotational direction.

[0153] Additionally, according to the elevator **1** of this embodiment, the drive motor **350** is attached to the bracket **51** so that the tip of the base face **350***a* aligns with the tip of the upper face **41***a* of the front protrusion part **41** of the support frame **4**. In other words, when attaching the drive motor **350**, the tip of the base face **350***a* and the tip of the upper face **41***a* are used as the reference for positioning the drive motor **350** in the front-rear direction. Therefore, it is possible to improve the workability in attaching the drive motor **350**.

[0154] Moreover, according to the elevator **1** of this embodiment, the brackets **51** and **54** to **56** supporting the drive motor **350**, the switch **354**, the limit switch **355**, and the limit switch **356** respectively have the upper faces **51***b* and **54***b* to **56***b* in a part and are attached to the support frame **4** so that the upper faces **51***b* and **54***b* to **56***b* are at the same height position as the upper face **41***a* of the front protrusion part **41** of the support frame **4**. In other words, when attaching the brackets **51** and **54** to **56**, the upper face **41***a* and the upper faces **51***b* and **54***b* to **56***b* are used as the reference for positioning the brackets **51** and **54** to **56** in the up-down direction. Therefore, it is possible to improve the workability in the attachment work.

[0155] Furthermore, according to the elevator **1** of this embodiment, the brackets **51** to **58** are attached to the support frame **4** using the fasteners **61** to **68** that can be fixed and released by a rotational operation from the front. Therefore, it is possible to improve the workability in the repair, replacement, and maintenance work when the need arises to repair or replace various components after constructing the elevator **1**.

[0156] Moreover, according to the elevator **1** of this embodiment, the fastener **61** of the bracket **51** supporting the drive motor **350** is a bolt and nut fastener, and the bracket **51** has the notched hole **51***a* that opens at the lower edge as the insertion hole for the shaft portion of the bolt **610**. Thus, the drive motor **350** and the bracket **51** can be lifted upward and removed in a state where the fastener **61** is loosened. Therefore, it is possible to improve the workability in the attachment and removal work.
[0157] Additionally, according to the elevator **1** of this embodiment, the fastener **62** of the bracket **52** supporting the idler pulley **352** is a square root bolt and nut fastener, and the support frame **4** and the bracket **52** respectively have the square hole **42***a* and the square hole **52***a* as the insertion holes for the shaft portion of the square root bolt **620**. The square root portion of the square root bolt **620** fits into the square hole **42***a* and the square hole **52***a*. Therefore, even if rotational torque is generated in the bracket **52** supporting the idler pulley **352** along with the driving of the drive motor **350**, it is possible to prevent the bracket **52** from causing misalignment in the rotational direction.
[0158] Moreover, according to the elevator **1** of this embodiment, the fasteners **64** to **66** of the brackets **54** to **56** supporting the switch **354**, the limit switch **355**, and the limit switch **356** are bolt and nut fasteners. The support frame **4** has the elongated holes **44***a* to **46***a* that are long in either the left-right direction or the up-down direction as the insertion holes for the shaft portions of the bolts **640** to **660**, and the brackets **54** to **56** have the elongated holes **54***a* to **56***a* that are long in the other direction as the insertion holes for the shaft portions of the bolts **640** to **660**. Therefore, it is possible to adjust or fine-tune the positions of the brackets **54** to **56** in the left-right direction and the up-down direction, and consequently, the positions of the switch **354**, the limit switch **355**, and the limit switch **356** in the left-right direction and the up-down direction.
[0159] Furthermore, according to the elevator **1** of this embodiment, the door hangers **311** of the car doors **31** respectively include the extension part **311***a* and the extension part **311***b* that extend upward, and the car doors **31** are connected to the drive belt **353** by the extension part **311***a* and the extension part **311***b*. Therefore, it is possible to reduce the number of components and reduce the manufacturing cost. In addition, as described above, since the height position of the drive belt **353** becomes closer to the door hanger **311**, it is possible to shorten the extension length of the extension part **311***a* and the extension part **311***b*. Therefore, there is no need to take measures such as increasing the thickness of the material to enhance the rigidity of the extension part **311***a* and the extension part **311***b*, and it is possible to achieve weight reduction for the car door **31**.
[0160] In addition, according to the elevator **1** of this embodiment, the extension part **311***a* and the extension part **311***b* include bent portions that protrude rearward or forward, and the drive belt **353** is fixed using a fastener in a state of being in contact with the lower face or upper face of the bent portion. Therefore, it is possible to simplify the connecting structure between the car door **31** and the drive belt **353**.
[0161] Furthermore, according to the elevator **1** of this embodiment, the door hanger **311** includes a pair of first members provided to be spaced apart in the left-right direction, and the extension part **311***a* and the extension part **311***b* are formed on the left half portion or the right half portion divided by the center line in the left-right direction of one of the first members. Therefore, it is possible to manufacture and obtain two first members from one material, thereby improving the yield.
[0162] Moreover, according to the elevator **1** of this embodiment, the extension part **311***b* includes the detection piece **311***c* for the limit switch **355** and the limit switch **356**. Therefore, it is possible to reduce the number of components and reduce the manufacturing cost.
[0163] Furthermore, according to the elevator **1** of this embodiment, as shown in FIG. **22**A to FIG. **22**D, the bracket **50** functions as a base for the support frame **4** by grounding the lower end **50***a* of the main body part **500** and the tip **50***b* of the rear protrusion part **501** on the floor of the building, thereby lifting the support frame **4** off the floor. Therefore, in the installation work of the door

opening/closing device **35**, it is possible to reduce the consideration for damage to various components of the door opening/closing device **35**, such as the door rail **33**, the drive motor **350**, the drive pulley **351**, the idler pulley **352**, and the switches **354** to **356**.

[0164] Moreover, according to the elevator **1** of this embodiment, the rear protrusion part **501** in the bracket **50** is provided to protrude rearward from the upper portion of the main body part **500**. Thus, the support frame **4** is inclined so that the support frame **4** is more separated from the floor toward the upper side. Therefore, various components arranged in the open space above the support frame **4** are sufficiently separated from the floor, making it possible to reliably avoid undesirable situations such as hitting the floor.

[0165] Additionally, according to the elevator **1** of this embodiment, the bracket **50** has the length H**50** in the up-down direction that is longer than the length H**4** of the support frame **4** in the up-down direction. Thus, the lower end **50***a* and the tip **50***b* of the bracket **50** reliably make contact with the floor of the building. Therefore, the integrated structure can be placed on the floor in a stable state.

[0166] Furthermore, according to the elevator **1** of this embodiment, the bracket **50** is attached to the support frame **4** so as to be changeable in position in the up-down direction. Therefore, it is possible to change the position appropriately for both uses: as a base and as a bracket for fixing the integrated structure to the car **3**.

[0167] Moreover, according to the elevator **1** of this embodiment, the bracket **50** is attached to the support frame **4** using the fastener **60**, and the fastener **60** is a bolt and nut fastener. The bracket **50** has the slide elongated hole **500***a* that is long in the up-down direction as the insertion hole for the shaft portion of the bolt **600**. Therefore, it is possible to change the position by sliding the bracket **50** without removing the fastener **60**.

[0168] Furthermore, according to the elevator **1** of this embodiment, the slide elongated hole **500***a* is formed to be longer than the length H**4** of the support frame **4** in the up-down direction. Therefore, it is possible to increase the sliding amount of the bracket **50**.

[0169] Additionally, according to the elevator **1** of this embodiment, the bracket **50** includes the engaged part **502** at a location on the rear protrusion part **501** that is behind the center of gravity of the integrated structure. Thus, as shown in FIG. **23**, when the integrated structure is suspended by a suspension tool **7** at the engaged part **502**, the support frame **4** is inclined with the lower side positioned toward the rear. In this state, when the integrated structure is lowered, the shaft portion of the bolt **600**A of the fastener **60**A attached to the lower bracket **50**A becomes easier to be inserted into the notched hole **40***b* of the support frame **4**. Therefore, in the installation work of the door opening/closing device **35**, the work of assembling the integrated structure to the car **3** can be performed smoothly.

[0170] It should be noted that the disclosure is not limited to the embodiment described above, and various modifications are possible within the scope that does not deviate from the essence of the disclosure. And the operation and effect of the present invention are not limited to the above embodiments. That is, the embodiments disclosed herein should be considered to be illustrative in all respects and not restrictive. The scope of the present invention is indicated by the appended claims, not by the above description. It is also contemplated that the scope of the present invention includes all modifications within the meaning and scope of equivalence to the claims.

[0171] In the above embodiment, the door devices **30** and **22** are of the center-opening type (2CO) with one door respectively on the left and right sides. However, the disclosure is not limited thereto. It goes without saying that the disclosure is also applicable to a side-opening type (2S) door device with two doors, a center-opening type (4CO) door device with two doors respectively on the left and right sides, and a side-opening type (1S) door device with one door.

[0172] Further, in the above embodiment, the door rail **33** is provided separately from the support frame **4** and is attached to the front face of the support frame **4** via the legs **33***a*. However, the disclosure is not limited thereto. The door rail may be formed by appropriately bending the plate

material of the support frame and may be provided integrally with the support frame.
[0173] In the above embodiment, the first pulley is the drive pulley **351** and the second pulley **352** is the idler pulley. However, the disclosure is not limited thereto. The first pulley may be a reduction pulley that receives rotation transmitted from the drive motor via a reduction belt.
[0174] Additionally, in the above embodiment, the rear protrusion part **501** in the bracket **50** is formed by bending the upper portion of a single plate material rearward. However, the disclosure is not limited thereto. The rear protrusion part may be a separate component from the main body part and may be joined to the main body part by welding or the like. Besides, the position of the rear protrusion part is not limited to the upper edge of the main body part.
[0175] Moreover, as long as there is no physical interference, it is naturally possible to apply the technical elements described above to other configurations, replace the technical elements described above with other configurations, and combine the technical elements described above with each other, and this is, of course, intended by the disclosure.

Claims

1. A car of an elevator, comprising: a door rail, supporting a car door to be slidable in a left-right direction; a first pulley and a second pulley provided to be spaced apart in the left-right direction above the door rail, and a drive belt operatively connected to the car door and being wound around the first pulley and the second pulley; a door position detection part, detecting that the car door is at a predetermined position; and a support frame, comprising the door rail on a front face and brackets at a plurality of locations on an upper edge, and the support frame supporting the first pulley, the second pulley, and the door position detection part via the brackets, wherein the car door comprises a roller that engages with an upper edge of the door rail from above and rotates, and the upper edge of the support frame is positioned lower than an uppermost portion of the roller.
2. The car of the elevator according to claim 1, further comprising: a drive motor, directly connected to the first pulley, wherein the drive motor is supported on the support frame via one of the brackets.
3. The car of the elevator according to claim 1, wherein the brackets are attached to a rear face of a main body part of the support frame.
4. The car of the elevator according to claim 1, wherein the support frame is a member that is long in the left-right direction and is arranged along a vertical plane, and the first pulley, the second pulley, and the door position detection part are arranged in an open space between a front offset plane and a rear offset plane with reference to the vertical plane of the support frame, above the support frame.
5. The car of the elevator according to claim 4, further comprising: a cover that covers a part or entirety of the open space from front, rear, and above.
6. The car of the elevator according to claim 2, wherein the support frame comprises: a main body part; and a front protrusion part that protrudes forward from at least a part of an upper edge of the main body part, the bracket supporting the drive motor has an upper face in a part, and is attached to the support frame so that the upper face is at the same height position as an upper face of the front protrusion part, and the drive motor has a base face, and is attached to the bracket so that the base face straddles and contacts the upper face of the front protrusion part and the upper face of the bracket.
7. The car of the elevator according to claim 6, wherein the drive motor is attached to the bracket so that a tip of the base face aligns with a tip of the upper face of the front protrusion part.
8. The car of the elevator according to claim 1, wherein the support frame comprises: a main body part; and a front protrusion part that protrudes forward from at least a part of an upper edge of the main body part, and the brackets have an upper face in a part, and is attached to the support frame so that the upper face is at the same height position as an upper face of the front protrusion part.

9. The car of the elevator according to claim 1, wherein the brackets are attached to the support frame using a fastener that is able to be fixed and released by a rotational operation from front of the support frame.
10. The car of the elevator according to claim 1, wherein the car door comprises: a door panel; and a door hanger that is attached to an upper end portion of the door panel, the door hanger comprises an extension part where a portion of a main body part of the support frame protrudes and extends upward, and the car door is connected to the drive belt at the extension part.
11. The car of the elevator according to claim 10, wherein the extension part comprises a detection piece for the door position detection part.
12. The car of the elevator according to claim 1, wherein the support frame comprises a pair of brackets spaced apart in the left-right direction on a rear face, and each of the pair of brackets comprises: a main body part, attached to the support frame; and a rear protrusion part, protruding rearward from an upper portion of the main body part, wherein a lower end of the main body part and a tip of the rear protrusion part are grounded to function as a base of the support frame.
13. The car of the elevator according to claim 12, wherein each of the pair of brackets has a length in an up-down direction that is longer than a length of the support frame in the up-down direction.
14. The car of the elevator according to claim 12, wherein each of the pair of brackets is attached to the support frame so as to be changeable in position in the up-down direction.
15. The car of the elevator according to claim 12, wherein each of the pair of brackets comprises an engaged part for a suspension tool at a location on the rear protrusion part that is behind a center of gravity of an integrated structure of the door rail, the drive motor, the first pulley, the second pulley, and the support frame.
16. A car of an elevator, comprising: a door rail, supporting a car door to be slidable in a left-right direction; a first pulley and a second pulley provided to be spaced apart in the left-right direction above the door rail, and a drive belt operatively connected to the car door and being wound around the first pulley and the second pulley; a door position detection part, detecting that the car door is at a predetermined position; and a support frame, comprising the door rail on a front face and brackets at a plurality of locations on an upper edge, and the support frame supporting the first pulley, the second pulley, and the door position detection part via the brackets, wherein the brackets are attached to the support frame using a fastener that is able to be fixed and released by a rotational operation from front of the support frame.
17. An elevator, comprising the car of the elevator according to claim 1.